

The Lonely Runner Problem on Pair-Sum Lattices

A lossless finitization, the rigid-zone covering obstruction, and—culminating—a conditional scaling-closure theorem: the frontier cells of every rung close under three named hypotheses (the multi-fiber survivor inequality H_q —quantified in this revision—the volume law H_{1g} , and the resonance budget H_{2r}), with a ledger of proved, measured, and fitted inputs, intervals on all sampled estimates, and the open obligations reduced to named objects.

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Pair-Sum Reformulation Program

A structural paper: no solution, no new cases; the three hypotheses of the main theorem are open and are not silently established anywhere in the text. Audit revision (v2): the H_q repair, quantified hypotheses, scope levels fixed. Revision (v4): zonotope-strand record updated—the harmonic instance's deepest hole closed exactly ($p = 16/47$), the two $n = 6$ boundary rows resolved as strict refutations. Revision (v3): the last open certification row ($n = 5$, $m = 10$) closed at $7/22$; no theorem changed.